

Optimization

Optimization literature

Textbooks:

1. Nocedal J. and Wright S.J., *Numerical Optimization*, Springer Series in Operations Research, Springer, 636 pp, 1999.
2. Spall J.C., *Introduction to Stochastic Search and Optimization, Estimation, Simulation and Control*, Wiley, 595 pp, 2003.
3. Chong E.K.P. and Zak S.H., *An Introduction to Optimization, Second Edition*, John Wiley & Sons, New York, 476 pp, 2001.
4. Rao S.S., *Engineering Optimization - Theory and Practice*, John Wiley & Sons, New York, 903 pp, 1996.
5. Gill P.E., Murray W. and Wright M.H., *Practical Optimization*, Elsevier, 401 pp., 2004.
6. Goldberg D.E., *Genetic Algorithms in Search, Optimization and Machine Learning*, Addison Wesley, Reading, Mass., 1989.
7. S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.(available at <http://www.stanford.edu/~boyd/cvxbook/>)

Optimization

Course Schedule:

1. Introduction to Optimization
2. Classical Optimization Techniques
3. Linear programming and the Simplex method
4. Nonlinear programming-One Dimensional Minimization Methods
5. Nonlinear programming-Unconstrained Optimization Techniques
6. Nonlinear programming-Constrained Optimization Techniques
7. Global Optimization Methods-Genetic algorithms
8. Global Optimization Methods-Simulated Annealing
9. Global Optimization Methods- Coupled Local Minimizers

1. Introduction

- Optimization is the act of obtaining the best result under given circumstances.
- Optimization can be defined as the process of finding the conditions that give the maximum or minimum of a function.
- The optimum seeking methods are also known as *mathematical programming techniques* and are generally studied as a part of operations research.
- *Operations research* is a branch of mathematics concerned with the application of scientific methods and techniques to decision making problems and with establishing the best or optimal solutions.

1. Introduction

- **Operations research** (in the UK) or **operational research (OR)** (in the US) is an interdisciplinary branch of mathematics which uses methods like:
 - mathematical modeling
 - statistics
 - algorithms to arrive at optimal or good decisions in complex problems which are concerned with optimizing the maxima (profit, faster assembly line, greater crop yield, higher bandwidth, etc) or minima (cost loss, lowering of risk, etc) of some objective function.
- The eventual intention behind using operations research is to elicit a best possible solution to a problem mathematically, which improves or optimizes the performance of the system.

1. Introduction

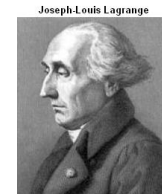
TABLE 1.1 Methods of Operations Research

Mathematical Programming Techniques	Stochastic Process Techniques	Statistical Methods
Calculus methods	Statistical decision theory	Regression analysis
Calculus of variations	Markov processes	Cluster analysis, pattern recognition
Nonlinear programming	Queueing theory	Design of experiments
Geometric programming	Renewal theory	Discriminate analysis (factor analysis)
Quadratic programming	Simulation methods	
Linear programming	Reliability theory	
Dynamic programming		
Integer programming		
Stochastic programming		
Separable programming		
Multiobjective programming		
Network methods: CPM and PERT		
Game theory		
Simulated annealing		
Genetic algorithms		
Neural networks		

1. Introduction

Historical development

- Isaac Newton (1642-1727)
(The development of differential calculus
methods of optimization)
- Joseph-Louis Lagrange (1736-1813)
(Calculus of variations, minimization of functionals,
method of optimization for constrained problems)
- Augustin-Louis Cauchy (1789-1857)
(Solution by direct substitution, steepest
descent method for unconstrained optimization)



1. Introduction

Historical development

- Leonhard Euler (1707-1783)
(Calculus of variations, minimization of functionals)
- Gottfried Leibnitz (1646-1716)
(Differential calculus methods of optimization)



Isim: Gottfried Wilhelm von Leibniz

1. Introduction

Historical development

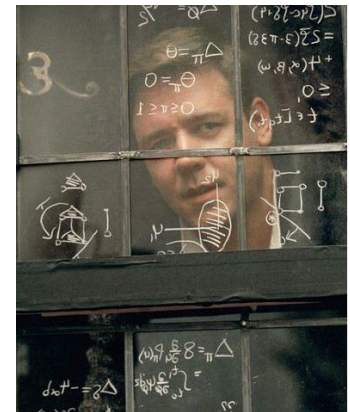
- George Bernard Dantzig (1914-2005)
(Linear programming and Simplex method (1947))
- Richard Bellman (1920-1984)
(Principle of optimality in dynamic programming problems)
- Harold William Kuhn (1925-)
(Necessary and sufficient conditions for the optimal solution of programming problems, game theory)



1. Introduction

Historical development

- Albert William Tucker (1905-1995)
(Necessary and sufficient conditions for the optimal solution of programming problems, nonlinear programming, game theory: his PhD student was John Nash)
- Von Neumann (1903-1957)
(game theory)



1. Introduction

- **Mathematical optimization problem:**

$$\text{minimize } f_0(x)$$

$$\text{subject to } g_i(x) \leq b_i, \quad i = 1, \dots, m$$

- $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$: objective function
- $x = (x_1, \dots, x_n)$: design variables (unknowns of the problem, they must be linearly independent)
- $g_i : \mathbf{R}^n \rightarrow \mathbf{R}$ ($i = 1, \dots, m$): inequality constraints
- The problem is a constrained optimization problem

1. Introduction

- If a point x^* corresponds to the minimum value of the function $f(x)$, the same point also corresponds to the maximum value of the negative of the function, $-f(x)$. Thus optimization can be taken to mean minimization since the maximum of a function can be found by seeking the minimum of the negative of the same function.

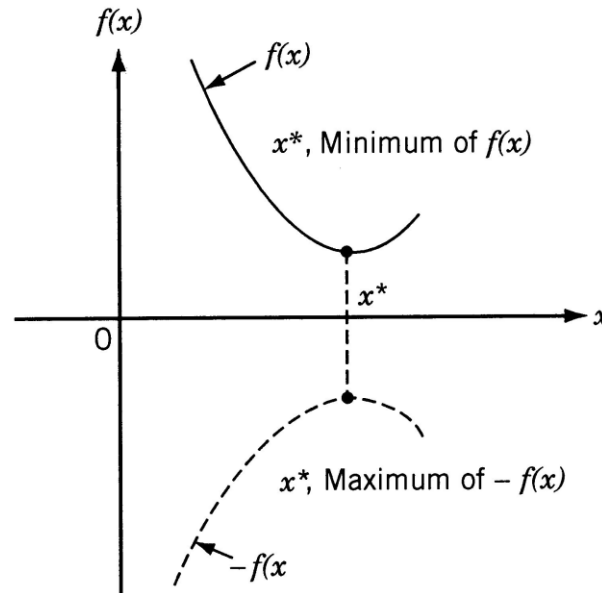


Figure 1.1 Minimum of $f(x)$ is same as maximum of $-f(x)$.

1. Introduction

Constraints

- **Behaviour constraints:** Constraints that represent limitations on the behaviour or performance of the system are termed *behaviour* or *functional constraints*.
- **Side constraints:** Constraints that represent physical limitations on design variables such as manufacturing limitations.

1. Introduction

- The conventional design procedures aim at finding an acceptable or adequate design which merely satisfies the functional and other requirements of the problem.
- In general, there will be more than one acceptable design, and the purpose of optimization is to choose the best one of the many acceptable designs available.
- Thus a criterion has to be chosen for comparing the different alternative acceptable designs and for selecting the best one.
- The criterion with respect to which the design is optimized, when expressed as a function of the design variables, is known as the **objective function**.

1. Introduction

- In civil engineering, the objective is usually taken as the minimization of the cost.
- In mechanical engineering, the maximization of the mechanical efficiency is the obvious choice of an objective function.
- In aerospace structural design problems, the objective function for minimization is generally taken as weight.
- In some situations, there may be more than one criterion to be satisfied simultaneously. An optimization problem involving multiple objective functions is known as a **multiobjective programming problem**.

1. Introduction

- With multiple objectives there arises a possibility of conflict, and one simple way to handle the problem is to construct an overall objective function as a linear combination of the conflicting multiple objective functions.
- Thus, if $f_1(\mathbf{X})$ and $f_2(\mathbf{X})$ denote two objective functions, construct a new (overall) objective function for optimization as:

$$f(\mathbf{X}) = \alpha_1 f_1(\mathbf{X}) + \alpha_2 f_2(\mathbf{X})$$

where α_1 and α_2 are constants whose values indicate the relative importance of one objective function to the other.

Examples

Design of civil engineering structures

- variables: width and height of member cross-sections
- constraints: limit stresses, maximum and minimum dimensions
- objective: minimum cost or minimum weight

Analysis of statistical data and building empirical models from measurements

- variables: model parameters
- Constraints: physical upper and lower bounds for model parameters
- Objective: prediction error

Classification of optimization problems

Classification based on:

- **Constraints**
 - Constrained optimization problem
 - Unconstrained optimization problem
- **Nature of the design variables**
 - Static optimization problems
 - Dynamic optimization problems

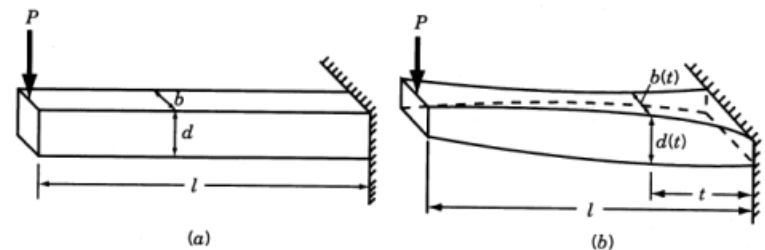
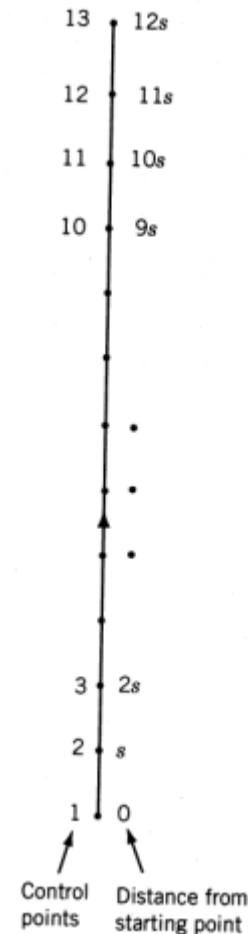


Figure 1.7 Cantilever beam under concentrated load.

Classification of optimization problems

Classification based on:

- **Physical structure of the problem**
 - Optimal control problems
 - Non-optimal control problems
- **Nature of the equations involved**
 - Nonlinear programming problem
 - Geometric programming problem
 - Quadratic programming problem
 - Linear programming problem



Classification of optimization problems

Classification based on:

- **Permissible values of the design variables**
 - Integer programming problems
 - Real valued programming problems
- **Deterministic nature of the variables**
 - Stochastic programming problem
 - Deterministic programming problem

Classification of optimization problems

Classification based on:

- **Separability of the functions**
 - Separable programming problems
 - Non-separable programming problems
- **Number of the objective functions**
 - Single objective programming problem
 - Multiobjective programming problem

Geometric Programming

- A geometric programming problem (GMP) is one in which the objective function and constraints are expressed as posynomials in $\mathbf{X}$.

Geometric programming

monomial function

$$f(x) = cx_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}, \quad \text{dom } f = \mathbf{R}_{++}^n$$

with $c > 0$; exponent a_i can be any real number

posynomial function: sum of monomials

$$f(x) = \sum_{k=1}^K c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}}, \quad \text{dom } f = \mathbf{R}_{++}^n$$

geometric program (GP)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 1, \quad i = 1, \dots, m \\ & h_i(x) = 1, \quad i = 1, \dots, p \end{array}$$

with f_i posynomial, h_i monomial

Quadratic Programming Problem

- A quadratic programming problem is a nonlinear programming problem with a quadratic objective function and linear constraints. It is usually formulated as follows:

$$F(\mathbf{X}) = c + \sum_{i=1}^n q_i x_i + \sum_{i=1}^n \sum_{j=1}^n Q_{ij} x_i x_j$$

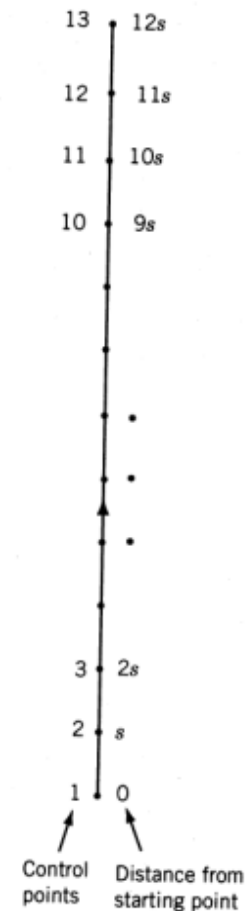
subject to

$$\sum_{i=1}^n a_{ij} x_i = b_j, \quad j = 1, 2, \dots, m$$
$$x_i \geq 0, \quad i = 1, 2, \dots, n$$

where c , q_i , Q_{ij} , a_{ij} , and b_j are constants.

Optimal Control Problem

- An optimal control (OC) problem is a mathematical programming problem involving a number of stages, where each stage evolves from the preceding stage in a prescribed manner.
- It is usually described by two types of variables: the control (design) and the state variables. The control variables define the system and govern the evolution of the system from one stage to the next, and the state variables describe the behaviour or status of the system in any stage.



Optimal Control Problem

- The problem is to find a set of control or design variables such that the total objective function (also known as the performance index) over all stages is minimized subject to a set of constraints on the control and state variables.

- An OC problem can be stated as follows:

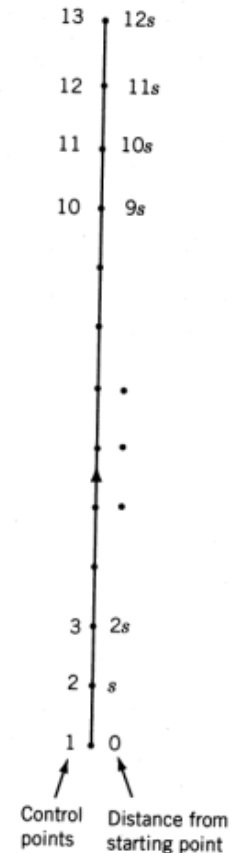
Find $\mathbf{X}$ which minimizes $f(\mathbf{X}) = \sum_{i=1}^l f_i(x_i, y_i)$
 subject to the constraints

$$q_i(x_i, y_i) + y_i = y_{i+1}, \quad i = 1, 2, \dots, l$$

$$\mathbf{g}_j(x_j) \leq 0, \quad j = 1, 2, \dots, l$$

$$\mathbf{h}_k(y_k) \leq 0, \quad k = 1, 2, \dots, l$$

where x_i is the i th control variable, y_i is the i th control variable, and f_i is the contribution of the i th stage to the total objective function; $\mathbf{g}_j$, $\mathbf{h}_k$ and q_i are functions of x_j , y_k and x_i and y_i , respectively, and l is the total number of stages.



Integer Programming Problem

- If some or all of the design variables $x_1, x_2, \dots, x_n$ of an optimization problem are restricted to take on only integer (or discrete) values, the problem is called an *integer programming problem*.
- If all the design variables are permitted to take any real value, the optimization problem is called a *real-valued programming problem*.

Stochastic Programming Problem

- A *stochastic programming problem* is an optimization problem in which some or all of the parameters (design variables and/or preassigned parameters) are probabilistic (nondeterministic or stochastic).
- In other words, stochastic programming deals with the solution of the optimization problems in which some of the variables are described by probability distributions.

Separable Programming Problem

- A function $f(x)$ is said to be *separable* if it can be expressed as the sum of n single variable functions, $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$, that is,

$$f(\mathbf{X}) = \sum_{i=1}^n f_i x_i$$

- A separable programming problem is one in which the objective function and the constraints are separable and can be expressed in standard form as:

Find $\mathbf{X}$ which minimizes $f(\mathbf{X}) = \sum_{i=1}^n f_i(x_i)$

subject to

$$g_j(\mathbf{X}) = \sum_{i=1}^n g_{ij}(x_i) \leq b_j, \quad j = 1, 2, \dots, m$$

where b_j is constant.

Multiojective Programming Problem

- A multiojective programming problem can be stated as follows:

Find $\mathbf{X}$ which minimizes $f_1(\mathbf{X}), f_2(\mathbf{X}), \dots, f_k(\mathbf{X})$
subject to

$$g_j(\mathbf{X}) \leq 0, \quad j = 1, 2, \dots, m$$

where $f_1, f_2, \dots, f_k$ denote the objective functions to be minimized simultaneously.

Review of mathematics

Concepts from linear algebra:

Positive definiteness

- **Test 1:** A matrix $\mathbf{A}$ will be positive definite if all its eigenvalues are positive; that is, all the values of λ that satisfy the determinantal equation

$$|\mathbf{A} - \lambda\mathbf{I}| = 0$$

should be positive. Similarly, the matrix $\mathbf{A}$ will be negative definite if its eigenvalues are negative.

Review of mathematics

Positive definiteness

- **Test 2:** Another test that can be used to find the positive definiteness of a matrix **A** of order n involves evaluation of the determinants

$$A_1 = |a_{11}|$$

$$A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$A_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$A_n = \begin{vmatrix} a_{11} & a_{12} & a_{13} \Lambda a_{1n} \\ a_{21} & a_{22} & a_{23} \Lambda a_{2n} \\ a_{31} & a_{32} & a_{33} \Lambda a_{3n} \\ M \\ a_{n1} & a_{n2} & a_{n3} \Lambda a_{nn} \end{vmatrix}$$

- The matrix **A** will be **positive definite** if and only if all the values $A_1, A_2, A_3, \dots, A_n$ are positive
- The matrix **A** will be **negative definite** if and only if the sign of A_j is $(-1)^j$ for $j=1,2,\dots,n$
- If some of the A_j are positive and the remaining A_j are zero, the matrix **A** will be **positive semidefinite**

Review of mathematics

Negative definiteness

- Equivalently, a matrix is **negative-definite** if all its **eigenvalues** are **negative**
- It is **positive-semidefinite** if all its **eigenvalues** are all **greater than or equal to zero**
- It is **negative-semidefinite** if all its **eigenvalues** are all **less than or equal to zero**

Review of mathematics

Concepts from linear algebra:

Nonsingular matrix: The determinant of the matrix is not zero.

Rank: The rank of a matrix $\mathbf{A}$ is the order of the largest nonsingular square submatrix of $\mathbf{A}$, that is, the largest submatrix with a determinant other than zero.

Review of mathematics

Solutions of a linear problem

Minimize $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$

Subject to $g(\mathbf{x})$: $\mathbf{Ax} = \mathbf{b}$

Side constraints: $\mathbf{x} \geq 0$

- The existence of a solution to this problem depends on the rows of $\mathbf{A}$.
- If the rows of $\mathbf{A}$ are linearly independent, then there is a unique solution to the system of equations.
- If $\det(\mathbf{A})$ is zero, that is, matrix $\mathbf{A}$ is singular, there are either *no solutions* or *infinite solutions*.

Review of mathematics

Suppose

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0.5 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} \quad \mathbf{A}^* = \begin{bmatrix} 1 & 1 & 1 & 3 \\ -1 & 1 & 0.5 & 1.5 \end{bmatrix}$$

The new matrix $\mathbf{A}^*$ is called the augmented matrix- the columns of $\mathbf{b}$ are added to $\mathbf{A}$. According to the theorems of linear algebra:

- If the augmented matrix $\mathbf{A}^*$ and the matrix of coefficients $\mathbf{A}$ have the same rank r which is less than the number of design variables n : ($r < n$), then there are many solutions.
- If the augmented matrix $\mathbf{A}^*$ and the matrix of coefficients $\mathbf{A}$ do not have the same rank, a solution does not exist.
- If the augmented matrix $\mathbf{A}^*$ and the matrix of coefficients $\mathbf{A}$ have the same rank $r=n$, where the number of constraints is equal to the number of design variables, then there is a unique solution.

Review of mathematics

In the example

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0.5 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} \quad \mathbf{A}^* = \begin{bmatrix} 1 & 1 & 1 & 3 \\ -1 & 1 & 0.5 & 1.5 \end{bmatrix}$$

The largest square submatrix is a 2 x 2 matrix (since $m = 2$ and $m < n$). Taking the submatrix which includes the first two columns of $\mathbf{A}$, the determinant has a value of 2 and therefore is nonsingular. Thus the rank of $\mathbf{A}$ is 2 ($r = 2$). The same columns appear in $\mathbf{A}^*$ making its rank also 2. Since $r < n$, infinitely many solutions exist.

Review of mathematics

In the example

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0.5 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} \quad \mathbf{A}^* = \begin{bmatrix} 1 & 1 & 1 & 3 \\ -1 & 1 & 0.5 & 1.5 \end{bmatrix}$$

One way to determine the solutions is to assign ($n-r$) variables arbitrary values and use them to determine values for the remaining r variables. The value $n-r$ is often identified as the *degree of freedom* for the system of equations.

In this example, the degree of freedom is 1 (i.e., $3-2$). For instance x_3 can be assigned a value of 1 in which case $x_1=0.5$ and $x_2=1.5$

2. Classical optimization techniques

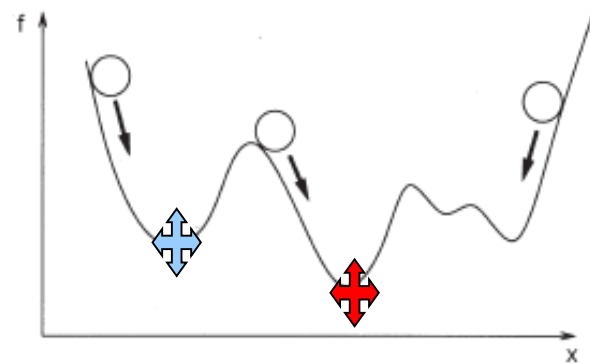
Single variable optimization

- Useful in finding the optimum solutions of continuous and differentiable functions
- These methods are analytical and make use of the techniques of differential calculus in locating the optimum points.
- Since some of the practical problems involve objective functions that are not continuous and/or differentiable, the classical optimization techniques have limited scope in practical applications.

2. Classical optimization techniques

Single variable optimization

- A function of one variable $f(x)$ has a relative or local minimum at $x = x^*$ if $f(x^*) \leq f(x^*+h)$ for all sufficiently small positive and negative values of h
- A point x^* is called a relative or local maximum if $f(x^*) \geq f(x^*+h)$ for all values of h sufficiently close to zero.



Global minima



Local minima

2. Classical optimization techniques

Single variable optimization

- A function $f(x)$ is said to have a global or absolute minimum at x^* if $f(x^*) \leq f(x)$ for all x , and not just for all x close to x^* , in the domain over which $f(x)$ is defined.
- Similarly, a point x^* will be a global maximum of $f(x)$ if $f(x^*) \geq f(x)$ for all x in the domain.

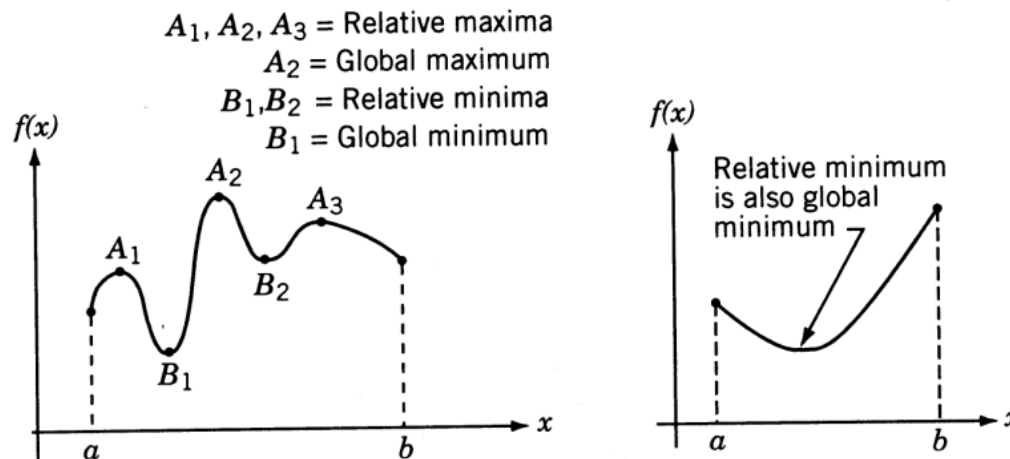
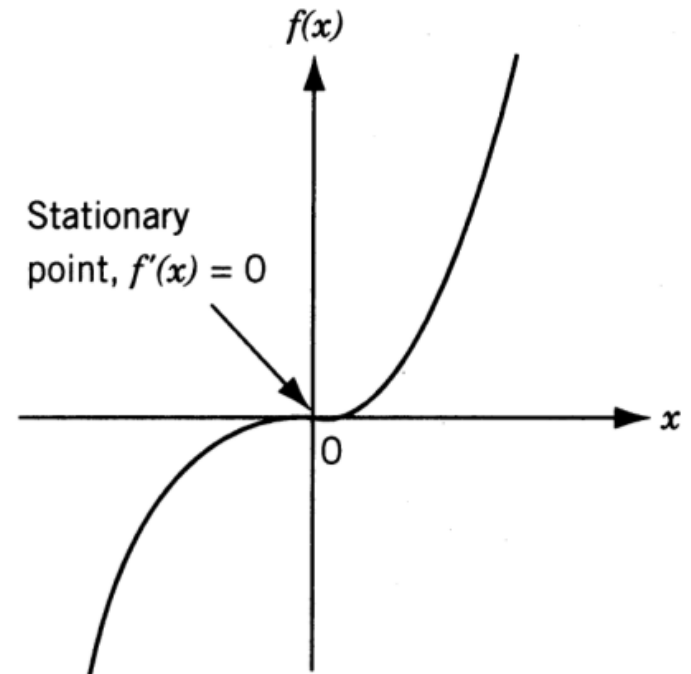


Figure 2.1 Relative and global minima.

Necessary condition

- If a function $f(x)$ is defined in the interval $a \leq x \leq b$ and has a relative minimum at $x = x^*$, where $a < x^* < b$, and if the derivative $df(x)/dx = f'(x)$ exists as a finite number at $x = x^*$, then $f'(x^*) = 0$
- The theorem does not say that the function necessarily will have a minimum or maximum at every point where the derivative is zero. *e.g.* $f'(x) = 0$ at $x = 0$ for the function shown in figure. However, this point is neither a minimum nor a maximum. In general, a point x^* at which $f'(x^*) = 0$ is called a *stationary point*.



Necessary condition

- The theorem does not say what happens if a minimum or a maximum occurs at a point x^* where the derivative fails to exist. For example, in the figure

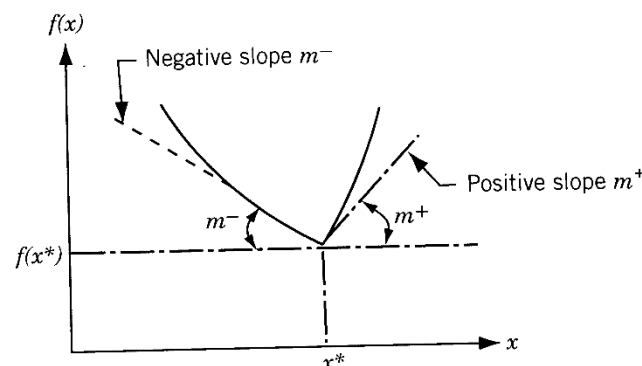


Figure 2.2 Derivative undefined at x^* .

$$\lim_{h \rightarrow 0} \frac{f(x^* + h) - f(x^*)}{h} = m^+ \text{ (positive) or } m^- \text{ (negative)}$$

depending on whether h approaches zero through positive or negative values, respectively. Unless the numbers m^+ or m^- are equal, the derivative $f'(x^*)$ does not exist. If $f'(x^*)$ does not exist, the theorem is not applicable.

Sufficient condition

- Let $f'(x^*)=f''(x^*)=\dots=f^{(n-1)}(x^*)=0$, but $f^{(n)}(x^*) \neq 0$. Then $f(x^*)$ is
 - A **minimum** value of $f(x)$ if $f^{(n)}(x^*) > 0$ and n is **even**
 - A **maximum** value of $f(x)$ if $f^{(n)}(x^*) < 0$ and n is **even**
 - Neither a minimum nor a maximum if n is **odd**

Example

Determine the maximum and minimum values of the function:

$$f(x) = 12x^5 - 45x^4 + 40x^3 + 5$$

Solution: Since $f'(x) = 60(x^4 - 3x^3 + 2x^2) = 60x^2(x-1)(x-2)$,
 $f'(x) = 0$ at $x=0, x=1$, and $x=2$.

The second derivative is:

$$f''(x) = 60(4x^3 - 9x^2 + 4x)$$

At $x=1$, $f''(x) = -60$ and hence $x=1$ is a relative maximum. Therefore,

$$f_{\max} = f(x=1) = 12$$

At $x=2$, $f''(x) = 240$ and hence $x=2$ is a relative minimum. Therefore,

$$f_{\min} = f(x=2) = -11$$

Example

Solution cont'd:

At $x=0$, $f''(x)=0$ and hence we must investigate the next derivative.

$$f'''(x) = 60(12x^2 - 18x + 4) = 240 \text{ at } x = 0$$

Since $f'''(x) \neq 0$ at $x=0$, $x=0$ is neither a maximum nor a minimum, and it is an inflection point.

Multivariable optimization with no constraints

- **Definition: *r*th Differential of *f***

If all partial derivatives of the function f through order $r \geq 1$ exist and are continuous at a point $\mathbf{X}^*$, the polynomial

$$d^r f(\mathbf{X}^*) = \underbrace{\sum_{i=1}^n \sum_{j=1}^n \wedge \sum_{k=1}^n h_i h_j \wedge h_k}_{r \text{ summations}} \frac{\partial^r f(\mathbf{X}^*)}{\partial x_i \partial x_j \wedge \partial x_k}$$

is called the r th differential of f at $\mathbf{X}^*$.

Multivariable optimization with no constraints

- **Example: *r*th Differential of *f***

$$d^r f(\mathbf{X}^*) = \underbrace{\sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n h_i h_j h_k}_{r \text{ summations}} \frac{\partial^r f(\mathbf{X}^*)}{\partial x_i \partial x_j \partial x_k}$$

when $r = 2$ and $n = 3$, we have

$$\begin{aligned} d^2 f(\mathbf{X}^*) &= d^2 f(x_1^*, x_2^*, x_3^*) = \sum_{i=1}^3 \sum_{j=1}^3 h_i h_j \frac{\partial^2 f(\mathbf{X}^*)}{\partial x_i \partial x_j} \\ &= h_1^2 \frac{\partial^2 f}{\partial x_1^2}(\mathbf{X}^*) + h_2^2 \frac{\partial^2 f}{\partial x_2^2}(\mathbf{X}^*) + h_3^2 \frac{\partial^2 f}{\partial x_3^2}(\mathbf{X}^*) \\ &\quad + 2h_1 h_2 \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{X}^*) + 2h_2 h_3 \frac{\partial^2 f}{\partial x_2 \partial x_3}(\mathbf{X}^*) + 2h_1 h_3 \frac{\partial^2 f}{\partial x_1 \partial x_3}(\mathbf{X}^*) \end{aligned}$$

Multivariable optimization with no constraints

- **Definition: *rth Differential of f***

The Taylor series expansion of a function $f(\mathbf{X}^*)$ about a point $\mathbf{X}^*$ is given by:

$$f(\mathbf{X}) = f(\mathbf{X}^*) + df(\mathbf{X}^*) + \frac{1}{2!} d^2 f(\mathbf{X}^*) + \frac{1}{3!} d^3 f(\mathbf{X}^*) \\ + \Lambda + \frac{1}{N!} d^N f(\mathbf{X}^*) + R_N(\mathbf{X}^*, \mathbf{h})$$

where the last term, called the remainder is given by:

$$R_N(\mathbf{X}^*, \mathbf{h}) = \frac{1}{(N+1)!} d^{N+1} f(\mathbf{X}^*, \theta \mathbf{h})$$

where $0 < \theta < 1$ and $\mathbf{h} = \mathbf{X} - \mathbf{X}^*$

Example

Find the second order Taylor's series approximation of the function

$$f(x_1, x_2, x_3) = x_2^2 x_3 + x_1 e^{x_3}$$

about the point

$$\mathbf{X}^* = \begin{Bmatrix} 1 \\ 0 \\ -2 \end{Bmatrix}$$

Solution: The second order Taylor's series approximation of the function f about point $\mathbf{X}^*$ is given by

$$f(\mathbf{X}) = f \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + df \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \frac{1}{2!} d^2 f \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

Example cont'd

where

$$f\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = e^{-2}$$

$$\begin{aligned} df\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} &= h_1 \frac{\partial f}{\partial x_1} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + h_2 \frac{\partial f}{\partial x_2} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + h_3 \frac{\partial f}{\partial x_3} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \\ &= [h_1 e^{x_3} + h_2 (2x_2 x_3) + h_3 x_2^2 + h_3 x_1 e^{x_3}] \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = h_1 e^{-2} + h_3 e^{-2} \end{aligned}$$

Example cont'd

where

$$\begin{aligned}
 d^2 f \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} &= \sum_{i=1}^3 \sum_{j=1}^3 h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = (h_1^2 \frac{\partial^2 f}{\partial x_1^2} + h_2^2 \frac{\partial^2 f}{\partial x_2^2} + h_3^2 \frac{\partial^2 f}{\partial x_3^2} \\
 &\quad + 2h_1 h_2 \frac{\partial^2 f}{\partial x_1 \partial x_2} + 2h_2 h_3 \frac{\partial^2 f}{\partial x_2 \partial x_3} + 2h_1 h_3 \frac{\partial^2 f}{\partial x_1 \partial x_3}) \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \\
 &= [h_1^2(0) + h_2^2(2x_3) + h_3^2(x_1 e^{x_3}) + 2h_1 h_2(0) + 2h_2 h_3(2x_2) + 2h_1 h_3(e^{x_3})] \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \\
 &= -4h_2^2 + e^{-2}h_3^2 + 2h_1 h_3 e^{-2}
 \end{aligned}$$

Example cont'd

Thus, the Taylor's series approximation is given by:

$$f(\mathbf{X}) \cong e^{-2} + e^{-2}(h_1 + h_3) + \frac{1}{2!}(-4h_2^2 + e^{-2}h_3^2 + 2h_1h_3e^{-2})$$

Where $h_1 = x_1 - 1$, $h_2 = x_2$, and $h_3 = x_3 + 2$

Multivariable optimization with no constraints

- **Necessary condition**

If $f(\mathbf{X})$ has an extreme point (maximum or minimum) at $\mathbf{X}=\mathbf{X}^*$ and if the first partial derivatives of $f(\mathbf{X})$ exist at $\mathbf{X}^*$, then

$$\frac{\partial f}{\partial x_1}(\mathbf{X}^*) = \frac{\partial f}{\partial x_2}(\mathbf{X}^*) = \Lambda = \frac{\partial f}{\partial x_n}(\mathbf{X}^*) = 0$$

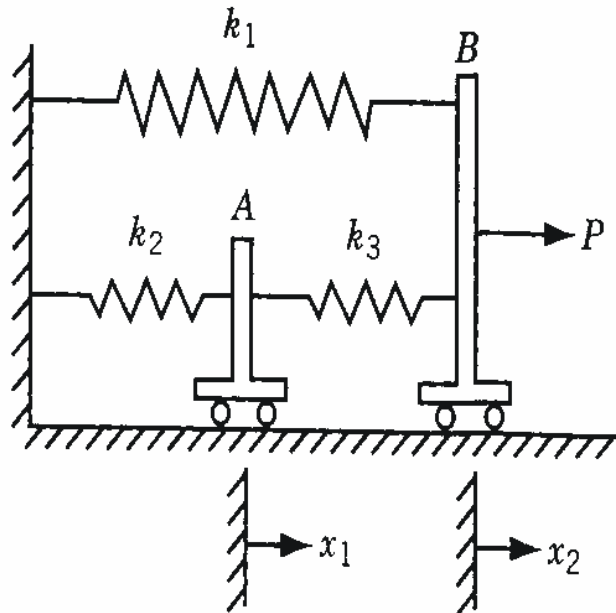
- **Sufficient condition**

A sufficient condition for a stationary point $\mathbf{X}^*$ to be an extreme point is that the matrix of second partial derivatives (Hessian matrix) of $f(\mathbf{X}^*)$ evaluated at $\mathbf{X}^*$ is

- **Positive definite** when $\mathbf{X}^*$ is a **relative minimum point**
- **Negative definite** when $\mathbf{X}^*$ is a **relative maximum point**

Example

Figure shows two frictionless rigid bodies (carts) A and B connected by three linear elastic springs having spring constants k_1 , k_2 , and k_3 . The springs are at their natural positions when the applied force P is zero. Find the displacements x_1 and x_2 under the force P by using the principle of minimum potential energy.



Example

Solution: According to the principle of minimum potential energy, the system will be in equilibrium under the load P if the potential energy is a minimum. The potential energy of the system is given by:

Potential energy (U)

= Strain energy of springs-work done by external forces

$$= \left[\frac{1}{2} k_2 x_1^2 + \frac{1}{2} k_3 (x_2 - x_1)^2 + \frac{1}{2} k_1 x_2^2 \right] - P x_2$$

The necessary condition for the minimum of U are

$$\left. \begin{aligned} \frac{\partial U}{\partial x_1} &= k_2 x_1 - k_3 (x_2 - x_1) = 0 \\ \frac{\partial U}{\partial x_2} &= k_3 (x_2 - x_1) + k_1 x_2 - P = 0 \end{aligned} \right\} \begin{aligned} x_1^* &= \frac{P k_3}{k_1 k_2 + k_1 k_3 + k_2 k_3} \\ x_2^* &= \frac{P (k_2 + k_3)}{k_1 k_2 + k_1 k_3 + k_2 k_3} \end{aligned}$$

Example

Solution cont'd: The sufficiency conditions for the minimum at (x_1^*, x_2^*) can also be verified by testing the positive definiteness of the Hessian matrix of U . The Hessian matrix of U evaluated at (x_1^*, x_2^*) is:

$$\mathbf{J} \Big|_{(x_1^*, x_2^*)} = \begin{bmatrix} \frac{\partial^2 U}{\partial x_1^2} & \frac{\partial^2 U}{\partial x_1 \partial x_2} \\ \frac{\partial^2 U}{\partial x_1 \partial x_2} & \frac{\partial^2 U}{\partial x_2^2} \end{bmatrix}_{(x_1^*, x_2^*)} = \begin{bmatrix} k_2 + k_3 & -k_3 \\ -k_3 & k_1 + k_3 \end{bmatrix}$$

The determinants of the square submatrices of $\mathbf{J}$ are

$$J_1 = |k_2 + k_3| = k_2 + k_3 > 0$$

$$J_2 = \begin{vmatrix} k_2 + k_3 & -k_3 \\ -k_3 & k_1 + k_3 \end{vmatrix} = k_1 k_2 + k_1 k_3 + k_2 k_3 > 0$$

Since the spring constants are always positive. Thus the matrix $\mathbf{J}$ is positive definite and hence (x_1^*, x_2^*) corresponds to the minimum of potential energy.

Semi-definite case

The sufficient conditions for the case when the Hessian matrix of the given function is semidefinite:

- In case of a function of a single variable, the higher order derivatives in the Taylor's series expansion are investigated

Semi-definite case

The sufficient conditions for a function of several variables for the case when the Hessian matrix of the given function is semidefinite:

- Let the partial derivatives of f of all orders up to the order $k \geq 2$ be continuous in the neighborhood of a stationary point $\mathbf{X}^*$, and

$$d^r f|_{\mathbf{X}=\mathbf{X}^*} = 0 \quad 1 \leq r \leq k-1$$

$$d^k f|_{\mathbf{X}=\mathbf{X}^*} \neq 0$$

so that $d^k f|_{\mathbf{X}=\mathbf{X}^*}$ is the first nonvanishing higher-order differential of f at $\mathbf{X}^*$.

- **If k is even:**
 - $\mathbf{X}^*$ is a relative minimum if $d^k f|_{\mathbf{X}=\mathbf{X}^*}$ is positive definite
 - $\mathbf{X}^*$ is a relative maximum if $d^k f|_{\mathbf{X}=\mathbf{X}^*}$ is negative definite
 - If $d^k f|_{\mathbf{X}=\mathbf{X}^*}$ is semidefinite, no general conclusions can be drawn
- **If k is odd, $\mathbf{X}^*$ is not an extreme point of $f(\mathbf{X}^*)$**

Saddle point

- In the case of a function of two variables $f(x,y)$, the Hessian matrix may be neither positive nor negative definite at a point (x^*,y^*) at which

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$$

- In such a case, the point (x^*,y^*) is called a saddle point.
- The characteristic of a saddle point is that it corresponds to a relative minimum or maximum of $f(x,y)$ wrt one variable, say, x (the other variable being fixed at $y=y^*$) and a relative maximum or minimum of $f(x,y)$ wrt the second variable y (the other variable being fixed at x^*).

Saddle point

Example: Consider the function
 $f(x,y)=x^2-y^2$. For this function:

$$\frac{\partial f}{\partial x} = 2x \text{ and } \frac{\partial f}{\partial y} = -2y$$

These first derivatives are zero at $x^* = 0$ and $y^* = 0$. The Hessian matrix of f at (x^*, y^*) is given by:

$$\mathbf{J} = \begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix}$$

Since this matrix is neither positive definite nor negative definite, the point $(x^*=0, y^*=0)$ is a saddle point.

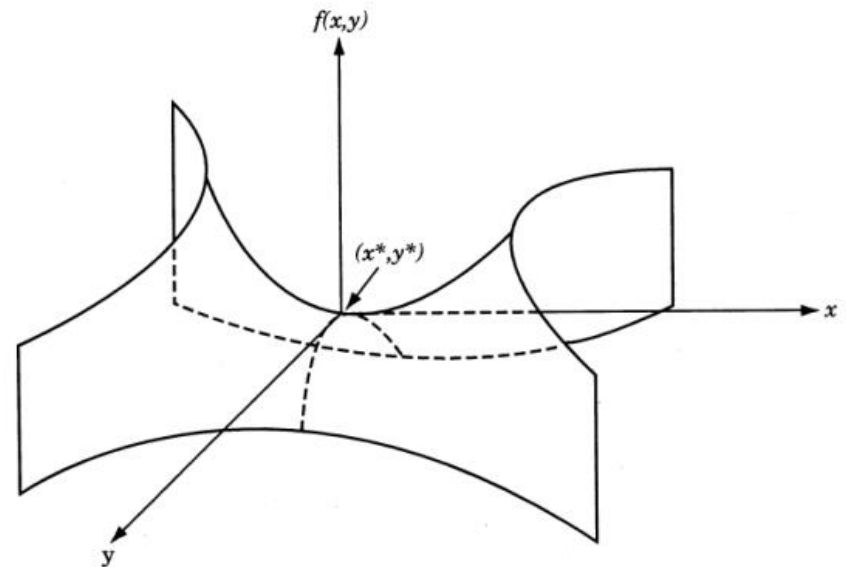


Figure 2.5 Saddle point of the function $f(x,y) = x^2 - y^2$.

Saddle point

Example cont'd:

It can be seen from the figure that $f(x, y^*) = f(x, 0)$ has a relative minimum and $f(x^*, y) = f(0, y)$ has a relative maximum at the saddle point (x^*, y^*) .

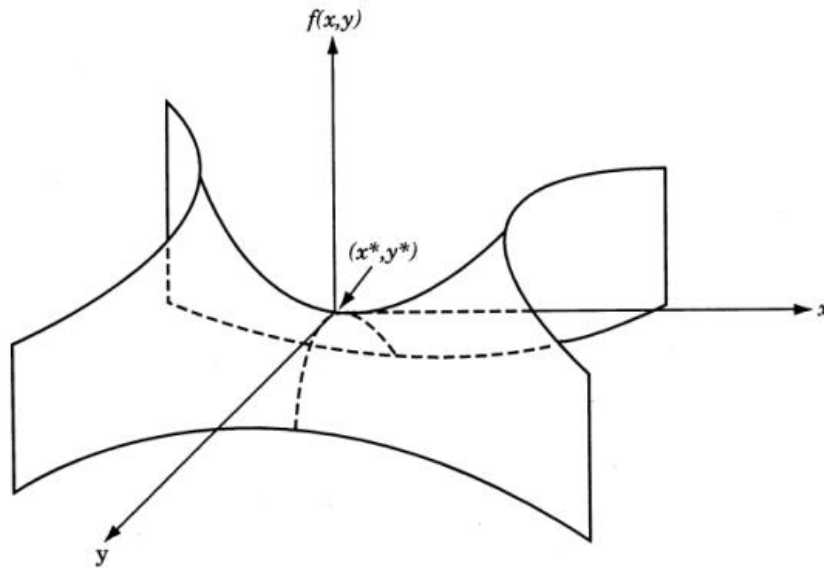


Figure 2.5 Saddle point of the function $f(x, y) = x^2 - y^2$.

Example

Find the extreme points of the function

$$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$$

Solution: The necessary conditions for the existence of an extreme point are:

$$\frac{\partial f}{\partial x_1} = 3x_1^2 + 4x_1 = x_1(3x_1 + 4) = 0$$

$$\frac{\partial f}{\partial x_2} = 3x_2^2 + 8x_2 = x_2(3x_2 + 8) = 0$$

These equations are satisfied at the points: (0,0), (0,-8/3), (-4/3,0), and (-4/3,-8/3)

Example

Solution cont'd: To find the nature of these extreme points, we have to use the sufficiency conditions. The second order partial derivatives of f are given by:

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1 + 4$$

$$\frac{\partial^2 f}{\partial x_2^2} = 6x_2 + 8$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$$

The Hessian matrix of f is given by:

$$\mathbf{J} = \begin{bmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{bmatrix}$$

Example

Solution cont'd:

$$\mathbf{J} = \begin{bmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{bmatrix}$$

If $J_1 = |6x_1 + 4|$ and $J_2 = \begin{vmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{vmatrix}$, the values of J_1 and J_2 and

the nature of the extreme point are as given in the next slide:

Example

Point $\mathbf{X}$	Value of $\mathbf{J}_1$	Value of $\mathbf{J}_2$	Nature of $\mathbf{J}$	Nature of $\mathbf{X}$	$f(\mathbf{X})$
(0,0)	+4	+32	Positive definite	Relative minimum	6
(0,-8/3)	+4	-32	Indefinite	Saddle point	418/27
(-4/3,0)	-4	-32	Indefinite	Saddle point	194/27
(-4/3,-8/3)	-4	+32	Negative definite	Relative maximum	50/3

Multivariable optimization with equality constraints

- **Problem statement:**

Minimize $f = f(\mathbf{X})$ subject to $g_j(\mathbf{X})=0, \quad j=1,2,\dots,m$ where

$$\mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$$

Here m is less than or equal to n , otherwise the problem becomes overdefined and, in general, there will be no solution.

- **Solution:**

- **Solution by direct substitution**
- **Solution by the method of constrained variation**
- **Solution by the method of Lagrange multipliers**

Solution by direct substitution

For a problem with n variables and m equality constraints:

- Solve the m equality constraints and express any set of m variables in terms of the remaining $n-m$ variables
- Substitute these expressions into the original objective function, the result is a new objective function involving only $n-m$ variables
- The new objective function is not subjected to any constraint, and hence its optimum can be found by using the unconstrained optimization techniques.

Solution by direct substitution

- Simple in theory
- Not convenient from a practical point of view as the constraint equations will be nonlinear for most of the problems
- Suitable only for simple problems

Example

Find the dimensions of a box of largest volume that can be inscribed in a sphere of unit radius

Solution: Let the origin of the Cartesian coordinate system x_1, x_2, x_3 be at the center of the sphere and the sides of the box be $2x_1, 2x_2$, and $2x_3$. The volume of the box is given by:

$$f(x_1, x_2, x_3) = 8x_1x_2x_3$$

Since the corners of the box lie on the surface of the sphere of unit radius, x_1, x_2 and x_3 have to satisfy the constraint

$$x_1^2 + x_2^2 + x_3^2 = 1$$

Example

This problem has three design variables and one equality constraint. Hence the equality constraint can be used to eliminate any one of the design variables from the objective function. If we choose to eliminate x_3 :

$$x_3 = (1 - x_1^2 - x_2^2)^{1/2}$$

Thus, the objective function becomes:

$$f(x_1, x_2) = 8x_1x_2(1 - x_1^2 - x_2^2)^{1/2}$$

which can be maximized as an unconstrained function in two variables.

Example

The necessary conditions for the maximum of f give:

$$\frac{\partial f}{\partial x_1} = 8x_2[(1 - x_1^2 - x_2^2)^{1/2} - \frac{x_1^2}{(1 - x_1^2 - x_2^2)^{1/2}}] = 0$$

$$\frac{\partial f}{\partial x_2} = 8x_1[(1 - x_1^2 - x_2^2)^{1/2} - \frac{x_2^2}{(1 - x_1^2 - x_2^2)^{1/2}}] = 0$$

which can be simplified as:

$$1 - 2x_1^2 - x_2^2 = 0$$

$$1 - x_1^2 - 2x_2^2 = 0$$

From which it follows that $x_1^* = x_2^* = 1/\sqrt{3}$ and hence $x_3^* = 1/\sqrt{3}$

Example

This solution gives the maximum volume of the box as:

$$f_{\max} = \frac{8}{3\sqrt{3}}$$

To find whether the solution found corresponds to a maximum or minimum, we apply the sufficiency conditions to $f(x_1, x_2)$ of the equation $f(x_1, x_2) = 8x_1x_2(1-x_1^2-x_2^2)^{1/2}$. The second order partial derivatives of f at (x_1^*, x_2^*) are given by:

$$\begin{aligned}\frac{\partial^2 f}{\partial x_1^2} &= -\frac{8x_1x_2}{(1-x_1^2-x_2^2)^{1/2}} - \frac{8x_2}{1-x_1^2-x_2^2} \left[\frac{x_1^3}{(1-x_1^2-x_2^2)^{1/2}} + 2x_1(1-x_1^2-x_2^2)^{1/2} \right] \\ &= -\frac{32}{\sqrt{3}} \text{ at } (x_1^*, x_2^*)\end{aligned}$$

Example

The second order partial derivatives of f at (x_1^*, x_2^*) are given by:

$$\begin{aligned}\frac{\partial^2 f}{\partial x_2^2} &= -\frac{8x_1x_2}{(1-x_1^2-x_2^2)^{1/2}} - \frac{8x_1}{1-x_1^2-x_2^2} \left[\frac{x_2^3}{(1-x_1^2-x_2^2)^{1/2}} + 2x_2(1-x_1^2-x_2^2)^{1/2} \right] \\ &= -\frac{32}{\sqrt{3}} \text{ at } (x_1^*, x_2^*)\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x_1 \partial x_2} &= 8(1-x_1^2-x_2^2)^{1/2} - \frac{8x_2^2}{(1-x_1^2-x_2^2)^{1/2}} \\ &\quad - \frac{8x_1^2}{1-x_1^2-x_2^2} \left[(1-x_1^2-x_2^2)^{1/2} + \frac{x_2^2}{(1-x_1^2-x_2^2)^{1/2}} \right] \\ &= -\frac{16}{\sqrt{3}} \text{ at } (x_1^*, x_2^*)\end{aligned}$$

Example

Since

$$\frac{\partial^2 f}{\partial x_2^2} < 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial x_1^2} \frac{\partial^2 f}{\partial x_2^2} - \left(\frac{\partial^2 f}{\partial x_1 \partial x_2} \right)^2 > 0$$

the Hessian matrix of f is negative definite at (x_1^*, x_2^*) .
Hence the point (x_1^*, x_2^*) corresponds to the maximum of f .

Solution by constrained variation

- Minimize $f(x_1, x_2)$
subject to $g(x_1, x_2) = 0$
- A necessary condition for f to have a minimum at some point (x_1^*, x_2^*) is that the total derivative of $f(x_1, x_2)$ wrt x_1 must be zero at (x_1^*, x_2^*)

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = 0$$

- Since $g(x_1^*, x_2^*) = 0$ at the minimum point, any variations dx_1 and dx_2 taken about the point (x_1^*, x_2^*) are called admissible variations provided that the new point lies on the constraint:

$$g(x_1^* + dx_1, x_2^* + dx_2) = 0$$

Solution by constrained variation

- Taylor's series expansion of the function about the point (x_1^*, x_2^*) :

$$g(x_1^* + dx_1, x_2^* + dx_2) \cong g(x_1^*, x_2^*) + \frac{\partial g}{\partial x_1}(x_1^*, x_2^*)dx_1 + \frac{\partial g}{\partial x_2}(x_1^*, x_2^*)dx_2 = 0$$

- Since $g(x_1^*, x_2^*) = 0$

$$dg = \frac{\partial g}{\partial x_1} dx_1 + \frac{\partial g}{\partial x_2} dx_2 = 0 \quad \text{at} \quad (x_1^*, x_2^*)$$

- Assuming $\frac{\partial g}{\partial x_2} \neq 0$

$$dx_2 = -\frac{\partial g / \partial x_1}{\partial g / \partial x_2}(x_1^*, x_2^*)dx_1$$

- Substituting the above equation into $df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = 0$

$$df = \left(\frac{\partial f}{\partial x_1} - \frac{\partial g / \partial x_1}{\partial g / \partial x_2} \frac{\partial f}{\partial x_2} \right) \Big|_{(x_1^*, x_2^*)} dx_1 = 0$$

Solution by constrained variation

$$df = \left(\frac{\partial f}{\partial x_1} - \frac{\partial g / \partial x_1}{\partial g / \partial x_2} \frac{\partial f}{\partial x_2} \right) \Big|_{(x_1^*, x_2^*)} dx_1 = 0$$

- The expression on the left hand side is called the *constrained variation* of f
- Since dx_1 can be chosen arbitrarily:

$$\left(\frac{\partial f}{\partial x_1} \frac{\partial g}{\partial x_2} - \frac{\partial f}{\partial x_2} \frac{\partial g}{\partial x_1} \right) \Big|_{(x_1^*, x_2^*)} = 0$$

- This equation represents a necessary condition in order to have (x_1^*, x_2^*) as an extreme point (minimum or maximum)

Example

A beam of uniform rectangular cross section is to be cut from a log having a circular cross section of diameter $2a$. The beam has to be used as a cantilever beam (the length is fixed) to carry a concentrated load at the free end. Find the dimensions of the beam that correspond to the maximum tensile (bending) stress carrying capacity.

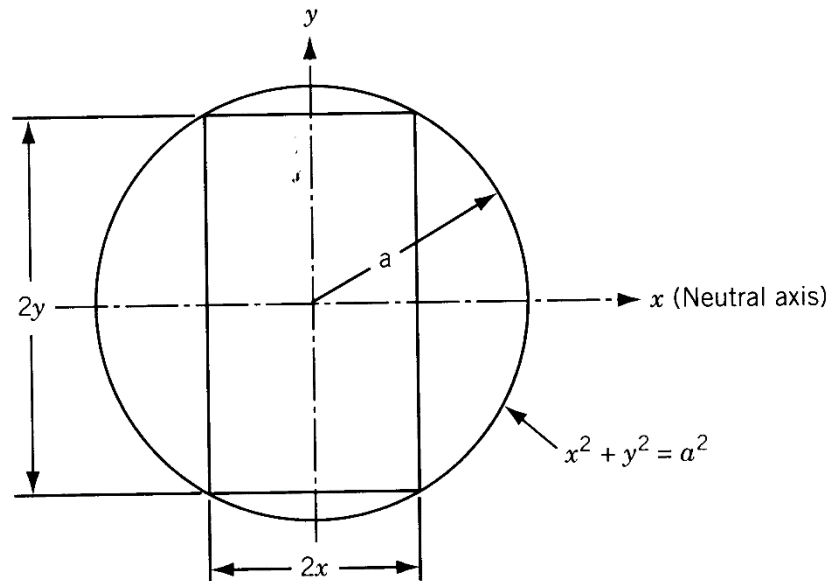


Figure 2.7 Cross section of the log.

Example

Solution: From elementary strength of materials, we know that the tensile stress induced in a rectangular beam σ at any fiber located at a distance y from the neutral axis is given by

$$\frac{\sigma}{y} = \frac{M}{I}$$

where M is the bending moment acting and I is the moment of inertia of the cross-section about the x axis. If the width and the depth of the rectangular beam shown in the figure are $2x$ and $2y$, respectively, the maximum tensile stress induced is given by:

$$\sigma_{\max} = \frac{M}{I} y = \frac{My}{\frac{1}{12} (2x)(2y)^3} = \frac{3}{4} \frac{M}{xy^2}$$

Example

solution cont'd: Thus for any specified bending moment, the beam is said to have maximum tensile stress carrying capacity if the maximum induced stress ($\sigma_{\max}$) is a minimum. Hence we need to minimize k/xy^2 or maximize Kxy^2 , where $k=3M/4$ and $K=1/k$, subject to the constraint

$$x^2 + y^2 = a^2$$

This problem has two variables and one constraint; hence the equation

$$\left(\frac{\partial f}{\partial x_1} \frac{\partial g}{\partial x_2} - \frac{\partial f}{\partial x_2} \frac{\partial g}{\partial x_1} \right) \Big|_{(x_1^*, x_2^*)} = 0$$

can be applied for finding the optimum solution.

Example

Solution: Since $f = kx^{-1}y^{-2}$
 $g = x^2 + y^2 - a^2$

$$\begin{aligned} \text{we have: } \frac{\partial f}{\partial x} &= -kx^{-2}y^{-2} & \frac{\partial g}{\partial x} &= 2x \\ \frac{\partial f}{\partial y} &= -2kx^{-1}y^{-3} & \frac{\partial g}{\partial y} &= 2y \end{aligned}$$

Equation $\left(\frac{\partial f}{\partial x_1} \frac{\partial g}{\partial x_2} - \frac{\partial f}{\partial x_2} \frac{\partial g}{\partial x_1}\right) \Big|_{(x_1^*, x_2^*)} = 0$ gives:

$$-kx^{-2}y^{-2}(2y) + 2kx^{-1}y^{-3}(2x) = 0 \quad \text{at} \quad (x^*, y^*)$$

Example

Solution: that is

$$y^* = \sqrt{2} x^*$$

Thus the beam of maximum tensile stress carrying capacity has a depth of $\sqrt{2}$ times its breadth. The optimum values of x and y can be obtained from the above equation and

$$g = x^2 + y^2 - a^2$$

as:

$$x^* = \frac{a}{\sqrt{3}} \text{ and } y^* = \sqrt{2} \frac{a}{\sqrt{3}}$$

Solution by constrained variation

Necessary conditions for a general problem

- The procedure described can be generalized to a problem with n variables and m constraints.
- In this case, each constraint equation $g_j(x)=0$, $j=1,2,\dots,m$ gives rise to a linear equation in the variations dx_i , $i=1,2,\dots,n$.
- Thus, there will be in all m linear equations in n variations. Hence any m variations can be expressed in terms of the remaining $n-m$ variations.
- These expressions can be used to express the differential of the objective function, df , in terms of the $n-m$ independent variations.
- By letting the coefficients of the independent variations vanish in the equation $df = 0$, one obtains the necessary conditions for the constrained optimum of the given function.

Solution by constrained variation

Necessary conditions for a general problem

- These conditions can be expressed as:

$$J\left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m}\right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \Lambda & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \Lambda & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \Lambda & \frac{\partial g_2}{\partial x_m} \\ \text{M} \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \Lambda & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \Lambda, n$$

- It is to be noted that the variations of the first m variables ($dx_1, dx_2, \dots, dx_m$) have been expressed in terms of the variations of the remaining $n-m$ variables ($dx_{m+1}, dx_{m+2}, \dots, dx_n$) in deriving the above equation.

Solution by constrained variation

Necessary conditions for a general problem

- This implies that the following relation is satisfied:

$$J\left(\frac{g_1, g_2, \Lambda, g_m}{x_1, x_2, \Lambda, x_m}\right) \neq 0$$

- The $n-m$ equations given by the below equation represent the necessary conditions for the extremum of $f(\mathbf{X})$ under the m equality constraints, $g_j(\mathbf{X}) = 0, j=1, 2, \dots, m$.

$$J\left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m}\right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \Lambda & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \Lambda & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \Lambda & \frac{\partial g_2}{\partial x_m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \Lambda & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \dots, n$$

Example

Minimize $f(\mathbf{Y}) = \frac{1}{2}(y_1^2 + y_2^2 + y_3^2 + y_4^2)$
 subject to
 $g_1(\mathbf{Y}) = y_1 + 2y_2 + 3y_3 + 5y_4 - 10 = 0$
 $g_2(\mathbf{Y}) = y_1 + 2y_2 + 5y_3 + 6y_4 - 15 = 0$

Solution: This problem can be solved by applying the necessary conditions given by

$$J \left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m} \right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \Lambda & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \Lambda & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \Lambda & \frac{\partial g_2}{\partial x_m} \\ \text{M} \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \Lambda & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \Lambda, n$$

Example

Solution cont'd: Since $n = 4$ and $m = 2$, we have to select two variables as independent variables. First we show that any arbitrary set of variables can not be chosen as independent variables since the remaining (dependent) variables have to satisfy the condition of

$$J\left(\frac{g_1, g_2, \Lambda, g_m}{x_1, x_2, \Lambda, x_m}\right) \neq 0$$

In terms of the notation of our equations, let us take the independent variables as $x_3=y_3$ and $x_4=y_4$ so that $x_1=y_1$ and $x_2=y_2$. Then the Jacobian becomes:

$$J\left(\frac{g_1, g_2}{x_1, x_2}\right) = \begin{vmatrix} \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_2} \\ \frac{\partial g_2}{\partial y_1} & \frac{\partial g_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 0$$

and hence the necessary conditions can not be applied.

Example

Solution cont'd: Next, let us take the independent variables as $x_3=y_2$ and $x_4=y_4$ so that $x_1=y_1$ and $x_2=y_3$. Then the Jacobian becomes:

$$J\left(\frac{g_1, g_2}{x_1, x_2}\right) = \begin{vmatrix} \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_3} \\ \frac{\partial g_2}{\partial y_1} & \frac{\partial g_2}{\partial y_3} \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 1 & 5 \end{vmatrix} = 2 \neq 0$$

and hence the necessary conditions of

$$J\left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m}\right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial \Lambda} & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial \Lambda} & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial \Lambda} & \frac{\partial g_2}{\partial x_m} \\ M & & & & \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \frac{\partial g_m}{\partial \Lambda} & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \Lambda, n$$

can be applied.

Example

Solution cont'd: The equation

$$J\left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m}\right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \Lambda & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \Lambda & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \Lambda & \frac{\partial g_2}{\partial x_m} \\ M & & & & \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \Lambda & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \Lambda, n$$

give for $k = m+1=3$

$$\begin{vmatrix} \frac{\partial f}{\partial x_3} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \\ \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{vmatrix} = \begin{vmatrix} \frac{\partial f}{\partial y_2} & \frac{\partial f}{\partial y_1} & \frac{\partial f}{\partial y_3} \\ \frac{\partial g_1}{\partial y_2} & \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_3} \\ \frac{\partial g_2}{\partial y_2} & \frac{\partial g_2}{\partial y_1} & \frac{\partial g_2}{\partial y_3} \end{vmatrix} = \begin{vmatrix} y_2 & y_1 & y_3 \\ 2 & 1 & 3 \\ 2 & 1 & 5 \end{vmatrix} = y_2(5-3) - y_1(10-6) + y_3(2-2) = 2y_2 - 4y_1 = 0$$

Example

Solution cont'd:

For $k = m+2=4$

$$\begin{vmatrix} \frac{\partial f}{\partial x_4} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \\ \frac{\partial g_1}{\partial x_4} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_4} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{vmatrix} = \begin{vmatrix} \frac{\partial f}{\partial y_4} & \frac{\partial f}{\partial y_1} & \frac{\partial f}{\partial y_3} \\ \frac{\partial g_1}{\partial y_4} & \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_3} \\ \frac{\partial g_2}{\partial y_4} & \frac{\partial g_2}{\partial y_1} & \frac{\partial g_2}{\partial y_3} \end{vmatrix} = \begin{vmatrix} y_4 & y_1 & y_3 \\ 5 & 1 & 3 \\ 6 & 1 & 5 \end{vmatrix} = y_4(5-3) - y_1(25-18) + y_3(5-6) = 2y_4 - 7y_1 - y_3 = 0$$

From the two previous equations, the necessary conditions for the minimum or the maximum of f is obtained as:

$$y_1 = \frac{1}{2} y_2$$

$$y_3 = 2y_4 - 7y_1 = 2y_4 - \frac{7}{2} y_2$$

Example

Solution cont'd:

When the equations

$$y_1 = \frac{1}{2} y_2$$

$$y_3 = 2y_4 - 7y_1 = 2y_4 - \frac{7}{2} y_2$$

are substituted, the equations

$$g_1(\mathbf{Y}) = y_1 + 2y_2 + 3y_3 + 5y_4 - 10 = 0$$

$$g_2(\mathbf{Y}) = y_1 + 2y_2 + 5y_3 + 6y_4 - 15 = 0$$

take the form:

$$-8y_2 + 11y_4 = 10$$

$$-15y_2 + 16y_4 = 15$$

Example

Solution cont'd:

from which the desired optimum solution can be obtained as:

$$y_1^* = -\frac{5}{74}$$

$$y_2^* = -\frac{5}{37}$$

$$y_3^* = \frac{155}{74}$$

$$y_4^* = \frac{30}{37}$$

Solution by constrained variation

Sufficiency conditions for a general problem

- By eliminating the first m variables, using the m equality constraints, the objective function can be made to depend only on the remaining variables $x_{m+1}, x_{m+2}, \dots, x_n$. Then the Taylor's series expansion of f , in terms of these variables, about the extreme point $\mathbf{X}^*$ gives:

$$f(\mathbf{X}^* + d\mathbf{X}) \cong f(\mathbf{X}^*) + \sum_{i=m+1}^n \left(\frac{\partial f}{\partial x_i} \right)_g dx_i + \frac{1}{2} \sum_{i=m+1}^n \sum_{j=m+1}^n \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_g dx_i dx_j$$

where $(\partial f / \partial x_i)_g$ is used to denote the partial derivative of f wrt x_i (holding all the other variables $x_{m+1}, x_{m+2}, \dots, x_{i-1}, x_{i+1}, x_{i+2}, \dots, x_n$ constant) when $x_1, x_2, \dots, x_m$ are allowed to change so that the constraints $g_j(\mathbf{X}^* + d\mathbf{X}) = 0, j = 1, 2, \dots, m$ are satisfied; the second derivative $(\partial^2 f / \partial x_i \partial x_j)_g$ is used to denote a similar meaning.

Solution by constrained variation

Example

Consider the problem of minimizing $f(\mathbf{X}) = f(x_1, x_2, x_3)$

Subject to the only constraint $g_1(\mathbf{X}) = x_1^2 + x_2^2 + x_3^2 - 8 = 0$

Since $n=3$ and $m=1$ in this problem, one can think of any of the m variables, say x_1 , to be dependent and the remaining $n-m$ variables, namely x_2 and x_3 , to be independent.

Here the constrained partial derivative $(\partial f / \partial x_2)_g$ means the rate of change of f with respect to x_2 (holding the other independent variable x_3 constant) and at the same time allowing x_1 to change about $\mathbf{X}^*$ so as to satisfy the constraint $g_1(\mathbf{X})=0$

Solution by constrained variation

Example

In the present case, this means that dx_1 has to be chosen to satisfy the relation

$$g_1(\mathbf{X}^* + d\mathbf{X}) \cong g_1(\mathbf{X}^*) + \frac{\partial g_1}{\partial x_1}(\mathbf{X}^*) dx_1 + \frac{\partial g_1}{\partial x_2}(\mathbf{X}^*) dx_2 + \frac{\partial g_1}{\partial x_3}(\mathbf{X}^*) dx_3 = 0$$

that is

$$2x_1^* dx_1 + 2x_2^* dx_2 = 0$$

since $g_1(\mathbf{X}^*)=0$ at the optimum point and $dx_3=0$ (x_3 is held constant.)

Solution by constrained variation

Example

Notice that $(df/dx_i)_g$ has to be zero for $i=m+1, m+2, \dots, n$ since the dx_i appearing in the equation

$$f(\mathbf{X}^* + d\mathbf{X}) \cong f(\mathbf{X}^*) + \sum_{i=m+1}^n \left(\frac{\partial f}{\partial x_i} \right)_g dx_i + \frac{1}{2} \sum_{i=m+1}^n \sum_{j=m+1}^n \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_g dx_i dx_j$$

are all independent. Thus, the necessary conditions for the existence of constrained optimum at $\mathbf{X}^*$ can also be expressed as:

$$\left(\frac{\partial f}{\partial x_i} \right)_g = 0, \quad i = m+1, m+2, \dots, n$$

Solution by constrained variation

Example

It can be shown that the equations

$$\left(\frac{\partial f}{\partial x_i} \right)_g = 0, \quad i = m+1, m+2, \dots, n$$

are nothing but the equation

$$J \left(\frac{f, g_1, g_2, \Lambda, g_m}{x_k, x_1, x_2, x_3, \Lambda, x_m} \right) = \begin{vmatrix} \frac{\partial f}{\partial x_k} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \Lambda & \frac{\partial f}{\partial x_m} \\ \frac{\partial g_1}{\partial x_k} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \Lambda & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_k} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \Lambda & \frac{\partial g_2}{\partial x_m} \\ \text{M} \\ \frac{\partial g_m}{\partial x_k} & \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} \Lambda & \frac{\partial g_m}{\partial x_m} \end{vmatrix} = 0 \quad \text{where } k = m+1, m+2, \Lambda, n$$

Sufficiency conditions for a general problem

- A sufficient condition for $\mathbf{X}^*$ to be a constrained relative minimum (maximum) is that the quadratic form Q defined by

$$Q = \sum_{i=m+1}^n \sum_{j=m+1}^n \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_g dx_i dx_j$$

is positive (negative) for all nonvanishing variations dx_i and the matrix

$$\begin{bmatrix} \left(\frac{\partial^2 f}{\partial x_{m+1}^2} \right)_g & \left(\frac{\partial^2 f}{\partial x_{m+1} \partial x_{m+2}} \right)_g & \Lambda & \left(\frac{\partial^2 f}{\partial x_{m+1} \partial x_n} \right)_g \\ \text{M} & & & \\ \left(\frac{\partial^2 f}{\partial x_n \partial x_{m+1}} \right)_g & \left(\frac{\partial^2 f}{\partial x_n \partial x_{m+2}} \right)_g & \Lambda & \left(\frac{\partial^2 f}{\partial x_n^2} \right)_g \end{bmatrix}$$

has to be positive (negative) definite to have Q positive (negative) for all choices of dx_i

Solution by constrained variation

- The computation of the constrained derivatives in the sufficiency condition is difficult and may be prohibitive for problems with more than three constraints
- Simple in theory
- Difficult to apply since the necessary conditions involve evaluation of determinants of order $m+1$

Solution by Lagrange multipliers

Problem with two variables and one constraint:

Minimize $f(x_1, x_2)$

Subject to $g(x_1, x_2) = 0$

For this problem, the necessary condition was found to be:

$$\left(\frac{\partial f}{\partial x_1} - \frac{\partial f / \partial x_2}{\partial g / \partial x_2} \frac{\partial g}{\partial x_1} \right) \Big|_{(x_1^*, x_2^*)} = 0$$

By defining a quantity λ , called the *Lagrange multiplier* as:

$$\lambda = - \left(\frac{\partial f / \partial x_2}{\partial g / \partial x_2} \right) \Big|_{(x_1^*, x_2^*)}$$

Solution by Lagrange multipliers

Problem with two variables and one constraint:

Necessary conditions for the point (x_1^*, x_2^*) to be an extreme point

The problem can be rewritten as:

$$\left(\frac{\partial f}{\partial x_2} + \lambda \frac{\partial g}{\partial x_2} \right) \Big|_{(x_1^*, x_2^*)} = 0$$

$$\left(\frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1} \right) \Big|_{(x_1^*, x_2^*)} = 0$$

In addition, the constraint equation has to be satisfied at the extreme point:

$$g(x_1, x_2) \Big|_{(x_1^*, x_2^*)} = 0$$

Solution by Lagrange multipliers

Problem with two variables and one constraint:

- The derivation of the necessary conditions by the method of Lagrange multipliers requires that at least one of the partial derivatives of $g(x_1, x_2)$ be nonzero at an extreme point.
- The necessary conditions are more commonly generated by constructing a function L , known as the Lagrange function, as

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2) = 0$$

Solution by Lagrange multipliers

Problem with two variables and one constraint:

- By treating L as a function of the three variables x_1 , x_2 and λ , the necessary conditions for its extremum are given by:

$$\frac{\partial L}{\partial x_1}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_1}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_1}(x_1, x_2) = 0$$

$$\frac{\partial L}{\partial x_2}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_2}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_2}(x_1, x_2) = 0$$

$$\frac{\partial L}{\partial \lambda}(x_1, x_2, \lambda) = g(x_1, x_2) = 0$$

Example

Example: Find the solution using the Lagrange multiplier method.

$$\text{Minimize } f(x, y) = kx^{-1}y^{-2}$$

subject to

$$g(x, y) = x^2 + y^2 - a^2 = 0$$

Solution

The Lagrange function is

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y) = kx^{-1}y^{-2} + \lambda(x^2 + y^2 - a^2)$$

The necessary conditions for the minimum of $f(x, y)$

$$\frac{\partial L}{\partial x} = -kx^{-2}y^{-2} + 2x\lambda = 0$$

$$\frac{\partial L}{\partial y} = -2kx^{-1}y^{-3} + 2y\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - a^2 = 0$$

Example

Solution cont'd

which yield:

$$2\lambda = \frac{k}{x^3 y^2} = \frac{2k}{xy^4}$$

from which the relation $x^* = (1/\sqrt{2})y^*$ can be obtained. This relation along with

$\frac{\partial L}{\partial \lambda} = x^2 + y^2 + a^2 = 0$ gives the optimum solution as :

$$x^* = \frac{a}{\sqrt{3}} \text{ and } y^* = \sqrt{2} \frac{a}{\sqrt{3}}$$

Solution by Lagrange multipliers

Necessary conditions for a general problem:

Minimize $f(\mathbf{X})$

subject to

$$g_j(\mathbf{X}) = 0, \quad j=1, 2, \dots, m$$

The Lagrange function, L , in this case is defined by introducing one Lagrange multiplier λ_j for each constraint $g_j(\mathbf{X})$ as

$$L(x_1, x_2, \Lambda, x_n, \lambda_1, \lambda_2, \Lambda, \lambda_m) = f(\mathbf{X}) + \lambda_1 g_1(\mathbf{X}) + \lambda_2 g_2(\mathbf{X}) + \Lambda + \lambda_m g_m(\mathbf{X})$$

Solution by Lagrange multipliers

By treating L as a function of the $n+m$ unknowns, $x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_m$, the necessary conditions for the extremum of L , which also corresponds to the solution of the original problem are given by:

$$\frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad i = 1, 2, \dots, n$$

$$\frac{\partial L}{\partial \lambda_j} = g_j(\mathbf{X}) = 0, \quad j = 1, 2, \dots, m$$

The above equations represent $n+m$ equations in terms of the $n+m$ unknowns, x_i and λ_j

Solution by Lagrange multipliers

The solution:

$$\mathbf{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{Bmatrix} \quad \text{and} \quad \lambda^* = \begin{Bmatrix} \lambda_1^* \\ \lambda_2^* \\ \vdots \\ \lambda_m^* \end{Bmatrix}$$

The vector $\mathbf{X}^*$ corresponds to the relative constrained minimum of $f(\mathbf{X})$ (sufficient conditions are to be verified) while the vector λ^* provides the sensitivity information.

Solution by Lagrange multipliers

Sufficient Condition

A sufficient condition for $f(\mathbf{X})$ to have a constrained relative minimum at $\mathbf{X}^*$ is that the quadratic Q defined by

$$Q = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 L}{\partial x_i \partial x_j} dx_i dx_j$$

evaluated at $\mathbf{X}=\mathbf{X}^*$ must be positive definite for all values of $d\mathbf{X}$ for which the constraints are satisfied.

If

$$Q = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 L}{\partial x_i \partial x_j}(\mathbf{X}^*, \lambda^*) dx_i dx_j$$

is negative for all choices of the admissible variations dx_i , $\mathbf{X}^*$ will be a constrained maximum of $f(\mathbf{X})$

Solution by Lagrange multipliers

A necessary condition for the quadratic form Q to be positive (negative) definite for all admissible variations $d\mathbf{X}$ is that each root of the polynomial z_i , defined by the following determinantal equation, be positive (negative):

$$\begin{vmatrix} L_{11} - z & L_{12} & L_{13} & \cdots & L_{1n} & g_{11} & g_{21} & \cdots & g_{m1} \\ L_{21} & L_{22} - z & L_{23} & \cdots & L_{2n} & g_{12} & g_{22} & \cdots & g_{m2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & L_{n3} & \cdots & L_{nn} - z & g_{1n} & g_{2n} & \cdots & g_{mn} \\ g_{11} & g_{12} & g_{13} & \cdots & g_{1n} & 0 & 0 & \cdots & 0 \\ g_{21} & g_{22} & g_{23} & \cdots & g_{2n} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{m1} & g_{m2} & g_{m3} & \cdots & g_{mn} & 0 & 0 & \cdots & 0 \end{vmatrix} = 0 \quad (2.44)$$

where

$$L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j}(\mathbf{X}^*, \lambda^*) \quad (2.45)$$

$$g_{ij} = \frac{\partial g_i}{\partial x_j}(\mathbf{X}^*) \quad (2.46)$$

- The determinantal equation, on expansion, leads to an $(n-m)$ th-order polynomial in z . If some of the roots of this polynomial are positive while the others are negative, the point $\mathbf{X}^*$ is not an extreme point.

Example 1

Find the dimensions of a cylindrical tin (with top and bottom) made up of sheet metal to maximize its volume such that the total surface area is equal to $A_0 = 24\pi$.

Solution

If x_1 and x_2 denote the radius of the base and length of the tin, respectively, the problem can be stated as:

$$\text{Maximize } f(x_1, x_2) = \pi x_1^2 x_2$$

subject to

$$2\pi x_1^2 + 2\pi x_1 x_2 = A_0 = 24\pi$$

Example 1

Solution

Maximize $f(x_1, x_2) = \pi x_1^2 x_2$

subject to $2\pi x_1^2 + 2\pi x_1 x_2 = A_0 = 24\pi$

The Lagrange function is:

$$L(x_1, x_2, \lambda) = \pi x_1^2 x_2 + \lambda(2\pi x_1^2 + 2\pi x_1 x_2 - A_0)$$

and the necessary conditions for the maximum of f give:

$$\frac{\partial L}{\partial x_1} = 2\pi x_1 x_2 + 4\pi \lambda x_1 + 2\pi \lambda x_2 = 0$$

$$\frac{\partial L}{\partial x_2} = \pi x_1^2 + 2\pi \lambda x_1 = 0$$

$$\frac{\partial L}{\partial \lambda} = 2\pi x_1^2 + 2\pi x_1 x_2 - A_0 = 0$$

Example 1

Solution

$$\lambda = -\frac{x_1 x_2}{2x_1 + x_2} = -\frac{1}{2}x_1$$

that is,

$$x_1 = \frac{1}{2}x_2$$

The above equations give the desired solution as:

$$x_1^* = \left(\frac{A_0}{6\pi}\right)^{1/2}, \quad x_2^* = \left(\frac{2A_0}{3\pi}\right)^{1/2}, \text{ and } \lambda^* = -\left(\frac{A_0}{24\pi}\right)^{1/2}$$

Example 1

Solution

This gives the maximum value of f as $f^* = \left(\frac{A_0^3}{54\pi} \right)^{1/2}$

If $A_0 = 24\pi$, the optimum solution becomes

$$x_1^* = 2, \quad x_2^* = 4, \quad \lambda^* = -1, \text{ and } f^* = 16\pi$$

To see that this solution really corresponds to the maximum of f , we apply the sufficiency condition of equation

$$\begin{vmatrix} L_{11} - z & L_{12} & L_{13} & \cdots & L_{1n} & g_{11} & g_{21} & \cdots & g_{m1} \\ L_{21} & L_{22} - z & L_{23} & \cdots & L_{2n} & g_{12} & g_{22} & \cdots & g_{m2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & L_{n3} & \cdots & L_{nn} - z & g_{1n} & g_{2n} & \cdots & g_{mn} \\ g_{11} & g_{12} & g_{13} & \cdots & g_{1n} & 0 & 0 & \cdots & 0 \\ g_{21} & g_{22} & g_{23} & \cdots & g_{2n} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{m1} & g_{m2} & g_{m3} & \cdots & g_{mn} & 0 & 0 & \cdots & 0 \end{vmatrix} = 0 \quad (2.44)$$

where

$$L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j} (\mathbf{X}^*, \lambda^*) \quad (2.45)$$

$$g_{ij} = \frac{\partial g_i}{\partial x_j} (\mathbf{X}^*) \quad (2.46)$$

Example 1

Solution

In this case:

$$L_{11} = \frac{\partial^2 L}{\partial x_1^2} \Big|_{(x^*, \lambda^*)} = 2\pi x_2^* + 4\pi \lambda^* = 4\pi$$

$$L_{12} = \frac{\partial^2 L}{\partial x_1 \partial x_2} \Big|_{(x^*, \lambda^*)} = 2\pi x_1^* + 2\pi \lambda^* = 2\pi$$

$$L_{22} = \frac{\partial^2 L}{\partial x_2^2} \Big|_{(x^*, \lambda^*)} = 0$$

$$g_{11} = \frac{\partial g_1}{\partial x_1} \Big|_{(x^*, \lambda^*)} = 4\pi x_1^* + 2\pi x_2^* = 16\pi$$

$$g_{12} = \frac{\partial g_1}{\partial x_2} \Big|_{(x^*, \lambda^*)} = 2\pi x_1^* = 4\pi$$

Example 1

Solution

Thus, equation

$$\begin{vmatrix} L_{11} - z & L_{12} & L_{13} & \cdots & L_{1n} & g_{11} & g_{21} & \cdots & g_{m1} \\ L_{21} & L_{22} - z & L_{23} & \cdots & L_{2n} & g_{12} & g_{22} & \cdots & g_{m2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & L_{n3} & \cdots & L_{nn} - z & g_{1n} & g_{2n} & \cdots & g_{mn} \\ g_{11} & g_{12} & g_{13} & \cdots & g_{1n} & 0 & 0 & \cdots & 0 \\ g_{21} & g_{22} & g_{23} & \cdots & g_{2n} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{m1} & g_{m2} & g_{m3} & \cdots & g_{mn} & 0 & 0 & \cdots & 0 \end{vmatrix} = 0 \quad (2.44)$$

where

$$L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j} (\mathbf{X}^*, \lambda^*) \quad (2.45)$$

becomes

$$g_{ij} = \frac{\partial g_i}{\partial x_j} (\mathbf{X}^*) \quad (2.46)$$

$$\begin{vmatrix} 4\pi - z & 2\pi & 16\pi \\ 2\pi & 0 - z & 4\pi \\ 16\pi & 4\pi & 0 \end{vmatrix} = 0$$

Example 1

Solution

that is,

$$272\pi^2 z + 192\pi^3 = 0$$

This gives

$$z = -\frac{12}{17}\pi$$

Since the value of z is negative, the point (x_1^*, x_2^*) corresponds to the maximum of f .

Example 2

Find the maximum of the function $f(\mathbf{X}) = 2x_1 + x_2 + 10$ subject to $g(\mathbf{X}) = x_1^2 + 2x_2^2 = 3$ using the Lagrange multiplier method. Also find the effect of changing the right-hand side of the constraint on the optimum value of f .

Solution

The Lagrange function is given by:

$$L(\mathbf{X}, \lambda) = 2x_1 + x_2 + 10 + \lambda(3 - x_1 - 2x_2^2)$$

The necessary conditions for the solution of the problem are:

$$\frac{\partial L}{\partial x_1} = 2 - \lambda = 0$$

$$\frac{\partial L}{\partial x_2} = 1 - 4\lambda x_2 = 0$$

$$\frac{\partial L}{\partial \lambda} = 3 - x_1 - 2x_2^2 = 0$$

Example 2

Solution

The solution of the equation is:

$$\mathbf{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \end{Bmatrix} = \begin{Bmatrix} 2.97 \\ 0.13 \end{Bmatrix}$$
$$\lambda^* = 2$$

The application of the sufficiency condition yields:

$$\begin{vmatrix} L_{11} - z & L_{12} & g_{11} \\ L_{21} & L_{22} - z & g_{12} \\ g_{11} & g_{12} & 0 \end{vmatrix} = 0$$

$$\begin{vmatrix} -z & 0 & -1 \\ 0 & -4\lambda - z & -4x_2 \\ -1 & -4x_2 & 0 \end{vmatrix} = \begin{vmatrix} -z & 0 & -1 \\ 0 & -8 - z & -0.52 \\ -1 & -0.52 & 0 \end{vmatrix} = 0$$

Example 2

Solution

$$0.2704z + 8 + z = 0$$

$$z = -6.2972$$

Hence $\mathbf{X}^*$ will be a maximum of f with $f^* = f(\mathbf{X}^*) = 16.07$